Keenan Eikenberry

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Research Interests

- (Self-supervised) representation learning
- Topics at the intersection of machine learning and formal methods (e.g., neural-guided algorithms, code representations, provably constrained AI systems, etc.)

Current Position

2023 – Present
- **Postdoctoral Research Associate**, Dartmouth College
  Department of Mathematics, Hanover, NH
  Supervisors: Yoonsang Lee and Anne Gelb

Education

2019 – 2023
- **Ph.D. in Applied Mathematics**, Arizona State University
  Advisors: Doug Cochran and Shiwei Lan
  Diss.: *Bayesian Inference for Markov Kernels Valued in Wasserstein Spaces*

2014 – 2016
- **M.A. in Mathematics**, Arizona State University
  Advisors: John Quigg and Steven Kaliszewski
  Thesis: *Functorial Results for C^*-algebras of Higher-Rank Graphs*

2009 – 2014
- **B.S. in Mathematics**, Arizona State University
  B.A. in English Literature (minor in Philosophy), Arizona State University
  Summa cum laude, GPA: 4.0

Papers and Code

Articles and Preprints

1. **Eikenberry, K.**, Liu, L., & Lee, Y. (2025). Post-training augmentation invariance. *arXiv preprint arXiv:2505.11702. (Under review.)* `https://github.com/keenan-eikenberry/augmentation_invariance`.
2. Eikenberry, S. E., Mancuso, M., Iboi, E., Phan, T., **Eikenberry, K.**, Kuang, Y., … Gumel, A. B. (2020). To mask or not to mask: Modeling the potential for face mask use by the general public to curtail the covid-19 pandemic. *Infectious disease modelling*, *5*, 293–308.

Selected Talks

- Wasserstein Dependence and Representation Learning. Applied and Computational Mathematics Seminar, Dartmouth College. November 2024.
- Maximizing Multiview Wasserstein Dependence in Autoencoders. SIAM Mathematics of Data Science Conference, Atlanta, Georgia. October 2024.
- Markov Kernels Valued in Wasserstein Spaces. Online CodEx Seminar. July 2023.
- Stacks of C^*-Categories. Great Plains Operator Theory Symposium, Texas A&M University. May 2019.

Selected Talks (continued)

- Group Actions on Topological k-Graphs. C*-Algebra Seminar, Arizona State University. November 2016.
- k-Graphs from Cayley Graphs. Great Plains Operator Theory Symposium, University of Illinois at Urbana-Champaign. May 2016.
- Functoriality for Higher-Rank Graph Algebras. C*-Algebra Seminar, Arizona State University. March 2016.

Skills

Mathematics	Analysis, category theory, probability, optimal transport, partial differential equations, differential geometry, algebraic topology, abstract algebra, operator algebras
ML and AI	(Self-supervised) representation learning, neural-guided formal methods, transformer architectures, optimal transport-based methods, theory of deep learning
Programming	Python (expert in PyTorch), Julia